
HIPCC Documentation

Release 1.1.1

Advanced Micro Devices, Inc.

Feb 24, 2026

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Note

ROCm provides and supports multiple compilers as described in [ROCm compiler reference](#).

hipcc is a compiler driver utility that will call clang or nvcc, depending on target, and pass the appropriate include and library options for the target compiler and HIP infrastructure. C++ executable versions of hipcc and hipconfig compiler driver utilities are provided.

The HIPCC public repository is located at <https://github.com/ROCm/llvm-project/tree/amd-staging/amd/hipcc>

The documentation is structured as follows:

Installation

- *[Building and testing HIPCC](#)*
- *[HIPCC environment variables](#)*

How to

- *[Using HIPCC](#)*

To contribute to the documentation, refer to [Contributing to ROCm](#).

You can find licensing information on the [Licensing](#) page.

BUILDING AND TESTING HIPCC

To build the `hipcc` and `hipconfig` executables, use the following commands.

```
mkdir build
cd build

cmake ..

make -j
```

Note

The tools are created in the current build folder, and will need to be copied to `/opt/rocm/hip/bin` folder location.

1.1 Testing HIPCC

Currently `hipcc` and `hipconfig` tools are tested by building and running test samples that can be found at [HIP-tests](#).

HIPCC ENVIRONMENT VARIABLES

This topic provides descriptions of the HIPCC environment variables. For more information about other ROCm environment variables, see [HIP environment variables](#).

Environment variable	Value
HIP_PLATFORM The platform targeted by HIP. If <code>HIP_PLATFORM</code> isn't set, then HIPCC attempts to auto-detect the platform based on whether the <code>nvcc</code> tool is found.	<code>amd, nvidia</code>
HIP_PATH The path of the HIP SDK on Microsoft Windows for AMD platforms.	Default: <code>C:/hip</code>
ROCM_PATH The path of the installed ROCm software stack on Linux for AMD platforms.	Default: <code>/opt/rocm</code>
CUDA_PATH Path to the CUDA SDK, which is only used for NVIDIA platforms.	Default: <code>/usr/local/cuda</code>
HIP_CLANG_PATH Path to the clang, which is only used for AMD platforms.	Default: <code>ROCM_PATH/llvm/bin</code> or <code>HIP_PATH/../llvm/bin</code>
HIP_LIB_PATH The HIP device library installation path.	Default: <code>HIP_PATH/lib</code>
HIP_DEVICE_LIB_PATH The HIP device library installation path.	
HIPCC_COMPILE_FLAGS_APPEND Append extra flags as compilation options to <code>hipcc</code> .	
HIPCC_LINK_FLAGS_APPEND Append extra flags as compilation options to <code>hipcc</code> .	
HIPCC_VERBOSE Outputs detailed information on subcommands executed during compilation.	1: Displays the command to <code>clang++</code> or <code>nvcc</code> with all options (<code>hipcc-cmd</code>). 2: Displays all relevant environment variables and their values. 4: Displays only the arguments passed to the <code>hipcc</code> command (<code>hipcc_args</code>). 5: Displays both the command to <code>clang++</code> or <code>nvcc</code> and <code>hipcc</code> arguments (<code>hipcc-cmd</code> and <code>hipcc_args</code>).

USING HIPCC

To use the newly built `hipcc` and `hipconfig` executables from the build folder use `./` in front of the executable name. For example:

```
./hipconfig --help  
./hipcc --help  
./hipcc --version  
./hipconfig --full
```


LICENSE

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